



AI-Driven Learning for Construction Challenges in Civil Engineering Design

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ABOUT

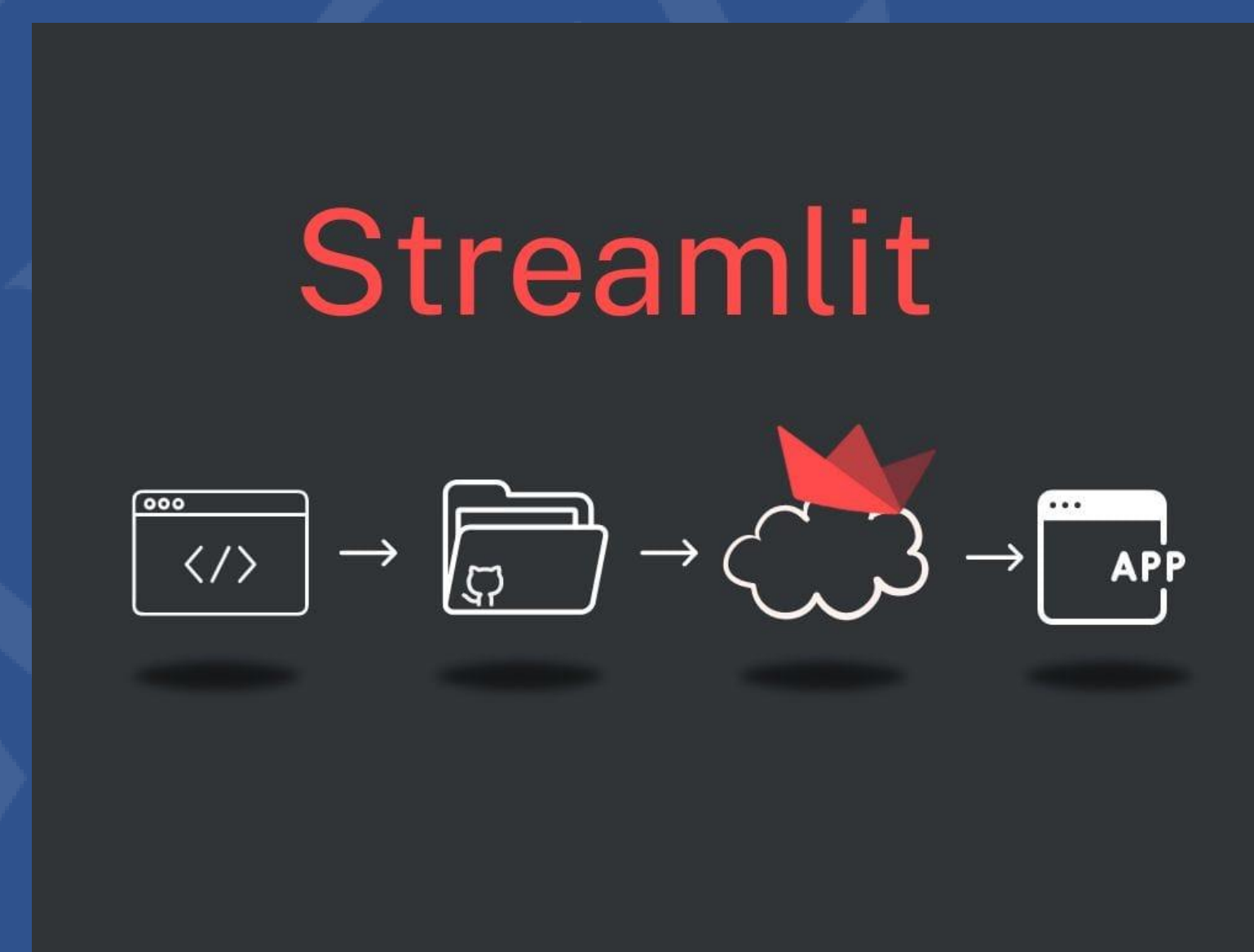
Need help with Professor Ahmed's course material? AnswerNest is here to assist! It's a tool built to help you quickly find answers to your questions using resources provided by Professor Ahmed himself.

AnswerNest has been provided personally some of Professor Ahmed's course textbooks, transcripts from selected YouTube videos, and other PDF documents. It uses a powerful Gemini AI model to understand your questions and provide relevant information, but more importantly, it **only** draws from these official course materials. This means you can be confident that the answers you receive are accurate and aligned with Professor Ahmed's teachings.

The website is built using Streamlit, a Python library that makes it easy to create interactive web apps. The backend entire project is written in Python.

We hope AnswerNest helps you succeed in your studies!

STREAMLIT



Code → Github → Streamlit → Website

CODE

```
def read_file(option):
    if option == "Chapter 11 - Deep Foundations":
        file = "Deep Foundation codotupd.pdf"
    elif option == "Shallow Foundation - Step by Step":
        file = "Shallow foundation step by step.pdf"
    elif option == "Types of Foundations Footings in Building Construction Video":
        file = "Types of Foundations Footings in Building Construction.pdf"
    elif option == "Shallow foundations and their types Foundations in Building Video":
        file = "Shallow Foundations and their types Foundations in Building 2.pdf"
    elif option == "Foundation Insulation Effectiveness Frost Protected Shallow Foundations Video":
        file = "Foundation Insulation Effectiveness Frost Protected Shallow Foundations.pdf"
    elif option == "Cold regions":
        file = "cold regionsREADABLE.pdf"
    elif option == "Foundations Issues":
        file = "Foundations IssuesREADABLE.pdf"
    elif option == "Dewatering":
        file = "dewateringREADABLE.pdf"

def generate(content):
    if user_input == "chat_input(something)":
        with st.chat_message("user"):
            st.session_state.messages.append(("user", content))

        prompt_text = ""
        """You are a helpful assistant. Your purpose is to answer questions based ONLY on the information provided in the attached document.

        If the answer to the question cannot be found directly within the document, you MUST respond with the phrase "Answer not found in document." Do not attempt to provide an answer based on your general knowledge, stick strictly to the content of the document.

        Document: {content}
        Question: {user_input}
        """

        api_key = st.session_state.get("api_key")
        client = GeminiClient(api_key=api_key)

        contents = [types.Content(role="user", partial=types.Part.from_text(text=prompt_text)), types.Part.from_text(text=content)]

        config = types.GenerateContentConfig(temperature=0, safety_settings=[types.SafetySetting(category="HARM_CATEGORY_HARASSMENT", threshold="BLOCK_MEDIUM_ABOVE"), types.SafetySetting(category="HARM_CATEGORY_SEXUALLY_EXPLICIT", threshold="BLOCK_MEDIUM_ABOVE"), types.SafetySetting(category="HARM_CATEGORY_HATE_SPEECH", threshold="BLOCK_MEDIUM_ABOVE"), types.SafetySetting(category="HARM_CATEGORY_DANGEROUS_CONTENT", threshold="BLOCK_MEDIUM_ABOVE")])

        with st.spinner("Generating..."):
            result = ""
            for chunk in client.models.generate_content_stream(
                contents, config=config, generation_config=config, stream=True
            ):
                result += chunk.text or ""

            response = result
            with st.chat_message("assistant"):
                st.session_state.messages.append(("assistant", response))
```

DOCUMENTS



- Step 1: Site Preparation**
- Clear vegetation, debris, and topsoil.
 - Establish site boundaries and layout the building plan using pegs, string lines, or total stations.
 - Conduct a final **geotechnical inspection** to confirm soil suitability.
- Possible Issues:**
- *Soft pockets or organic soil layers found:* Replace with compacted granular fill or lean concrete.
 - *Wrong layout:* Always verify layout with surveyor or project engineer before excavation.

- Step 2: Excavation**
- Excavate trenches or pits according to footing dimensions and depth as per drawings.
 - Maintain **vertical side walls** and proper **bottom levels**.
 - Use manual tools or machinery depending on the scale.
- Possible Issues:**
- *Over-excavation:* Fill back using compacted granular material or lean concrete to achieve the required level.
 - *Water ingress or seepage:* Use sump pumps or temporary drainage trenches. For high water table areas, consider well point dewatering.
 - *Sidewall collapse:* Use timber or metal shoring as needed.

- Textbooks
- Youtube Transcripts
- Any PDFs/TXTs

WEBSITE



<https://asif-ahmeds-ai-answerer.streamlit.app/>

GEMINI

Gemini 1.5 Flash-8B

Model code	models/gemini-1.5-flash-8b		
Supported data types	Inputs Audio, images, video, and text	Output Text	
Token limits ¹	Input token limit 1,048,576	Output token limit 8,192	
Audio/visual specs	Maximum number of images per prompt 3,000	Maximum video length 1 hour	Maximum audio length Approximately 9.5 hours
Capabilities	System instructions Supported	JSON mode Supported	JSON schema Supported
	Adjustable safety settings Supported	Caching Supported	Tuning Supported
	Function calling Supported	Code execution Supported	Live API Not supported
Versions	Read the model version patterns for more details. • Latest: gemini-1.5-flash-8b-latest • Latest stable: gemini-1.5-flash-8b • Stable: gemini-1.5-flash-8b-001		

QUESTIONS

Chapter 11 - Deep Foundations

- Suggest some critical items that a construction inspector should watch for during the construction of auger-cast piles
- Describe the differences between a drilled shaft and a driven pile

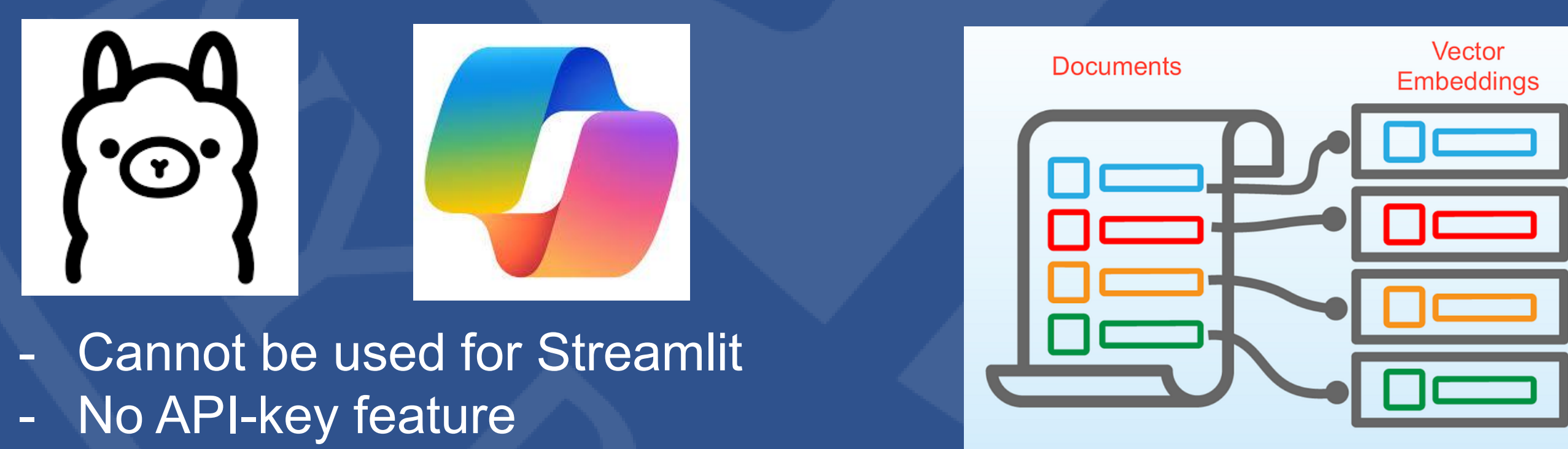
dewatering

- What is the purpose of a permanent dewatering system?
- When should a field pumping test be considered?

Types of Foundations Footings in Building Construction Video

- What are some types of shallow foundations?
- What factors play a role in selecting the suitable type of foundation?

CHALLENGES



- Cannot be used for Streamlit
- No API-key feature
- Switched to use Gemini

Gemma 3 27B
gemma-3-27b-it

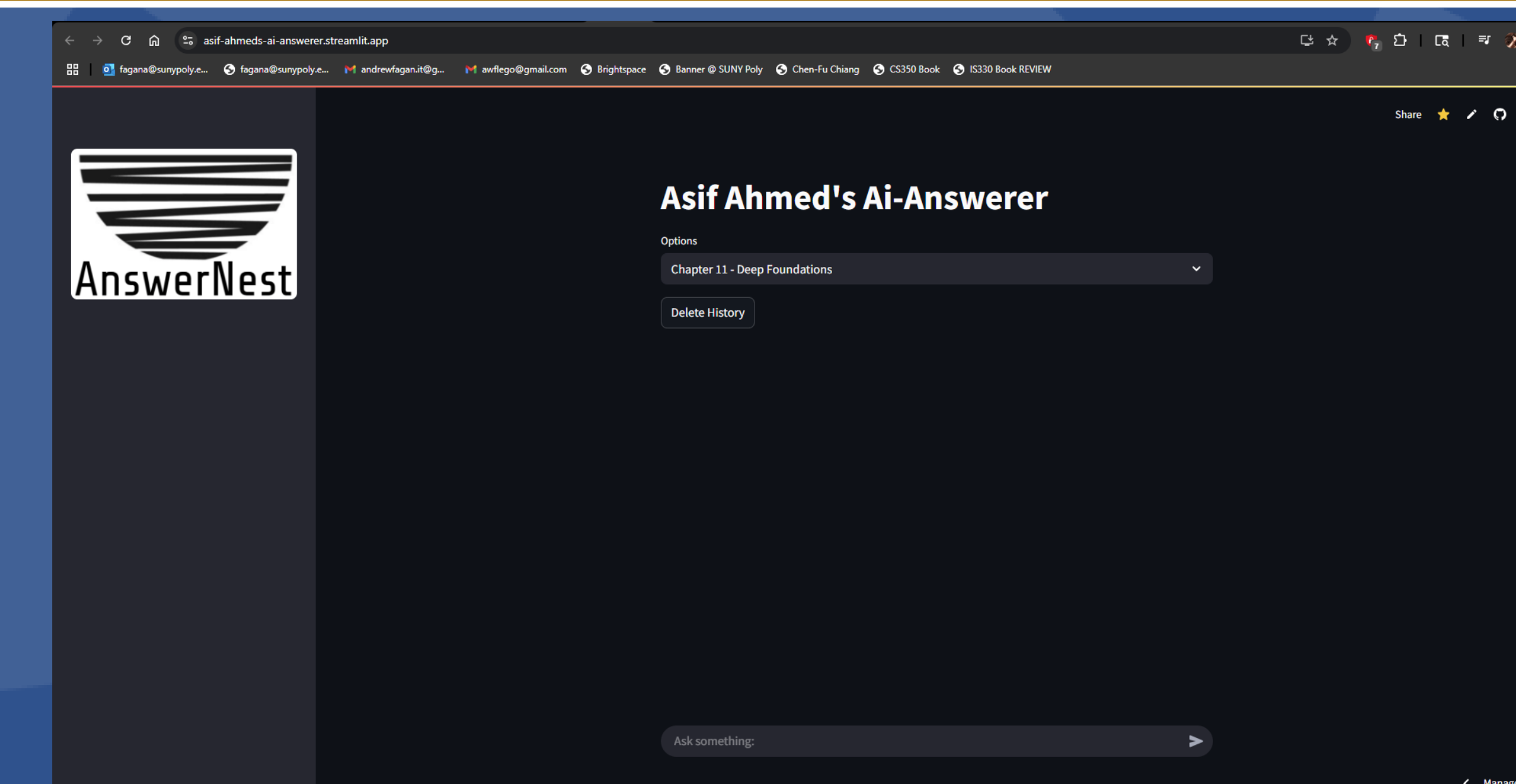
Rate limits

Free
30 RPM 14400 req/day

- Embedding vs Inheritance Models
- Unable to emulate it for this project
- Will continue to work on after the summer

- Insufficient Model
- Switched to gemini-1.5-8b

WEBSITE/GITHUB



Github Repository: https://github.com/AndrewFagan04/Sum25_fagana

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